Intelligent Agriculture Information System: State-Level

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Statement of Purpose

The purpose of this project is to develop an Intelligent Agriculture Information System for the state of Haryana powered by agentic Artificial Intelligence and Large Language Models (LLMs). By systematically assimilating diverse, multi-modal data—ranging from high-resolution satellite imagery and complex climatic models to unstructured policy documents and daily market prices—the system will construct a unified, geo-enabled knowledge graph. The ultimate objective is to deploy an AI agent capable of complex spatio-temporal reasoning to optimize state resource allocation, track the localized impact of agronomic interventions, mitigate ecological degradation, and enhance the economic viability of farming at the micro-geographic level.

Problem Framework

The structural architecture of this project is designed to solve four compounding crises within Haryana's agricultural ecosystem:

1. Ecological and Hydrological Degradation: The historical reliance on intensive rice-wheat cropping sequences has precipitated severe groundwater depletion and widespread soil salinization. With hundreds of thousands of acres suffering from waterlogging and degraded water tables, there is an urgent need for spatial intelligence to track soil health and model the effectiveness of reclamation efforts (such as targeted gypsum application).
2. Environmental Hazards via Crop Residue Burning: The post-harvest burning of paddy straw creates acute atmospheric pollution and destroys soil nutrients. While the state heavily subsidizes Crop Residue Management (CRM) machinery (like Super Seeders), policymakers currently lack the integrated tools to correlate hyper-local machinery deployment with satellite-derived thermal anomaly maps to verify if these subsidies are actually curbing burning events.
3. Economic Fragility and Policy Targeting: Farmers operate on incredibly tight profit margins driven by high variable costs and land rental rates. State financial safety nets, such as the Bhavantar Bharpai Yojana and crop diversification incentives, require highly accurate, real-time validations of crop acreage and health to properly disburse funds. Without precision yield modeling, state procurement liabilities and insurance claims cannot be reliably forecasted.
4. Data Siloing and Systemic Interoperability: The most significant technical bottleneck is the fracturing of critical intelligence. Highly accurate remote sensing data (GeoTIFFs), coarse evapotranspiration models, unstructured government policy PDFs (often in Hindi), and tabular market arrival feeds currently exist in isolated silos. This fragmentation—combined with varying spatial resolutions and metadata standards—prevents cohesive, data-driven agricultural planning and leads to misallocated resources.